PRATICAL-5

AIM:- Perform the Linear regression on the given data warehouse data using R/Python.

STEPS:-

Input Data

Step 1:- Values of height

151, 174, 138, 186, 128, 136, 179, 163, 152, 131

Step 2:-Values of weight.

63, 81, 56, 91, 47, 57, 76, 72, 62, 48

lm() Function :

This function creates the relationship model between the predictor and the response variable.

Syntax

Step 3:-The basic syntax for lm() function in linear regression is –

lm(formula,data)

Following is the description of the parameters used :–

- **formula is a symbol presenting the relation between x and y.**
- **data is the vector on which the formula will be applied.**

Step 4:-Create Relationship Model & get the Coefficients

```
# Values of height

x <- c(151, 174, 138, 186, 128, 136, 179, 163, 152, 131)

# Values of width

y <- c(63, 81, 56, 91, 47, 57, 76, 72, 62, 48)

# Apply the lm() function.

relation <- lm(y~x)

print(relation)
```

OUTPUT:

```
Call:
lm(formula = y ~ x)

Coefficients:
(Intercept)            x
   -38.4551       0.6746
```

Step 4:- Get the Summary of the Relationship

 # Values of height

x <- c(151, 174, 138, 186, 128, 136, 179, 163, 152, 131)

 # Values of width

y <- c(63, 81, 56, 91, 47, 57, 76, 72, 62, 48)

 # Apply the lm() function

relation <- lm(y~x)

print(summary(relation))

OUTPUT:

```
Call:
lm(formula = y ~ x)

Residuals:
    Min      1Q  Median      3Q     Max 
-6.3002 -1.6629  0.0412  1.8944  3.9775 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) -38.45509    8.04901  -4.778  0.00139 ** 
x             0.67461    0.05191  12.997 1.16e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.253 on 8 degrees of freedom
Multiple R-squared:  0.9548,	Adjusted R-squared:  0.9491 
F-statistic: 168.9 on 1 and 8 DF,  p-value: 1.164e-06
```

Step 5:- predict() Function

Syntax

predict(object, newdata)

Step 6:- Predict the weight of new persons

 # The predictor vector.

x <- c(151, 174, 138, 186, 128, 136, 179, 163, 152, 131)

 # The response vector.

y <- c(63, 81, 56, 91, 47, 57, 76, 72, 62, 48)

 # Apply the lm() function.

relation <- lm(y~x)

 # Find weight of a person with height 170.

a <- data.frame(x = 170)

result <- predict(relation,a)

print(result)

OUTPUT:

```
       1
76.22869
```

Step 6:- Visualize the Regression Graphically

 # Create the predictor and response variable

x <- c(151, 174, 138, 186, 128, 136, 179, 163, 152, 131)

y <- c(63, 81, 56, 91, 47, 57, 76, 72, 62, 48)

relation <- lm(y~x)

 # Give the chart file a name.

png(file = "linearregression.png")

 # Plot the chart.

plot(y,x,col = "blue",main = "Height & Weight Regression", abline(lm(x~y)),cex = 1.3,pch = 16,xlab = "Weight in Kg",ylab = "Height in cm")

 # Save the file.

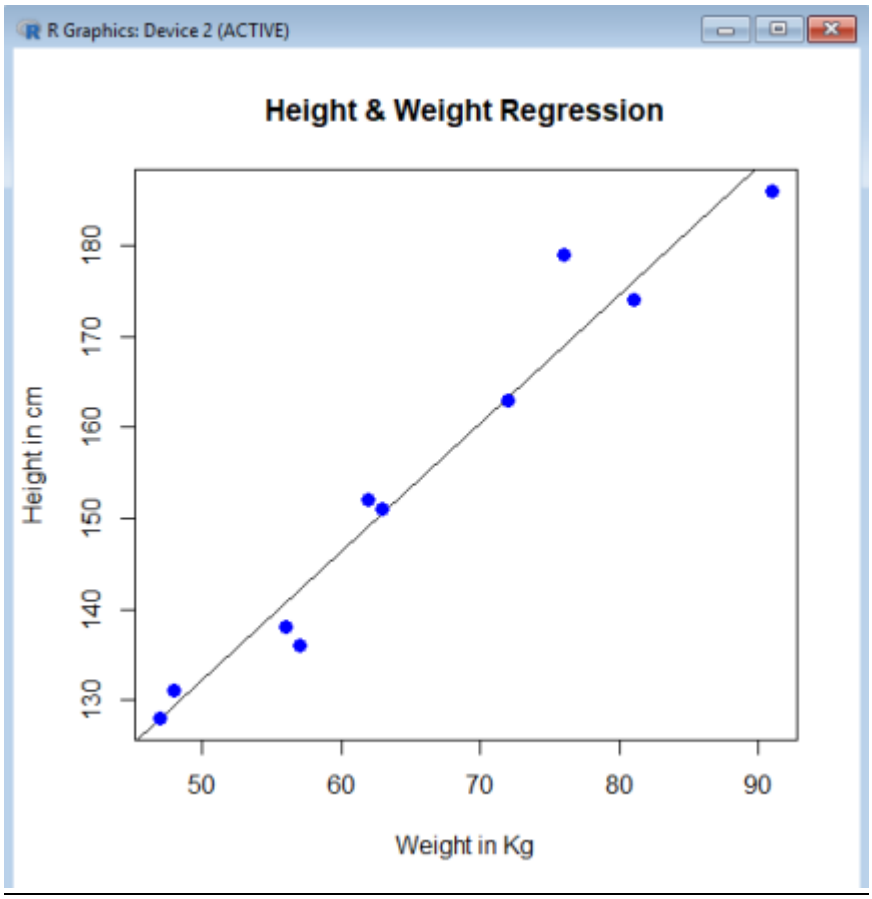
dev.off()

```
null device
          1
```

Step 7:- Plot the chart.

plot(y,x,col = "blue",main = "Height & Weight Regression", abline(lm(x~y)),cex = 1.3,pch = 16,xlab = "Weight in Kg",ylab = "Height in cm")

OUTPUT:


R Graphics: Device 2 (ACTIVE)
Height & Weight Regression
Height in cm
130
140
150
160
170
180
50
60
70
80
90
Weight in Kg